Running the PowerFactory model and the Python scripts

IEEE 13-node feeder, dual-setting directional recloser and fuse coordination
DIgSILENT PowerFactory 2021 SP2

This guide explains how to open the model in PowerFactory, how to read the main results from it, and how to run the Python scripts that repeat the complete study. Sections 3 to 6 are done by hand in the PowerFactory window. Section 7 runs the scripts, also from inside PowerFactory. Section 8 lists the problems that can come up.

1. What you need

PowerFactory 2021 SP2 with a working licence, and Python 3.9 (64-bit) on the same computer. PowerFactory uses the installed Python to run scripts. If Python 3.9 is not there, install it from `www.python.org` and tick "Add Python to PATH" on the first screen of the installer. An internet connection is needed once, in section 7.2.

2. Preparing the files

Copy the folder `1_Main_DSDR_Fuse_Coordination` to `C:\DSDR`. The scripts must end up in

`C:\DSDR\1_Main_DSDR_Fuse_Coordination\scripts`

because they look for each other there. Please do not rename or move anything inside the folder.

File	What it is
`01_MAIN_model_STANDIN_copy_DSDR_DG100pct.pfd`	the ready model (DG 100 %, dual-setting recloser, revised fuses)
`IEEE 13 Node Feeder.pfd`	the network without any protection; the model was built from it
`scripts\show_progress.py`	runs the whole study (section 7.3)
`scripts\install_packages.py`	installs two Python packages, once (section 7.2)

The time-domain models are in the folder `2_EMT_Fig10_time_domain`; they are only needed in section 6.

3. Opening the model

1. Start PowerFactory and log in.
2. Choose File, Import, Project (*.pfd), select `01_MAIN_model_STANDIN_copy_DSDR_DG100pct.pfd` and press Execute. The project is listed under your user as "IEEE13 DSDR penetration 100 %".
3. Right-click the project and choose Activate.
4. In the study case list at the top, choose **Study with Substation Transformer**.

Use this study case for all results. It contains the 5 MVA substation transformer, and with it the fault currents agree with the IEEE benchmark. In the other study case the source is stiff and the fault currents come out too high.

The single-line diagram shows the feeder. R1 is at the head of the feeder, R2 is on line 632–671, and the synchronous generator (DG) is at node 692.

4. Load flow without and with the DG

1. Double-click the Synchronous Machine at node 692, tick Out of Service and press OK.
2. Click Calculate Load Flow, choose AC load flow, unbalanced, 3-phase (ABC), and press Execute.
3. Read the current of line 650–632 (this is R1) and of line 632–671 (this is R2).

4. Take the tick away, so that the machine is in service again, and calculate once more.

	DG out	DG in
Line 650–632, recloser R1	587.7 A	284.9 A
Line 632–671, recloser R2	470.0 A forward	252.2 A reversed

With the DG, the current through R2 becomes smaller and flows in the opposite direction. That is the reason R2 gets a second, reverse setting.

5. Short circuit and tripping times

Keep the DG in service and do the following.

1. Right-click the busbar 633 and choose Calculate, Short-Circuit.
2. Set the method to Complete, the fault type to 3-phase short-circuit, the fault impedance to 0, and press Execute.
3. Read the current in fuse F633, in R1 and in R2.
4. Double-click the devices listed below and read their tripping time.

Three-phase fault at node 633, DG in	Current	Tripping time
R1 Fast (R1, fast curve)	3970 A	0.106 s
R2 Rev Fast (R2, reverse group, dual setting)	1500 A	0.060 s
R2 Fast (R2, forward group, single setting)	1500 A	0.160 s
F633 (fuse, starts to melt)	5426 A	0.149 s

The fuse is saved when the recloser trips before the fuse starts to melt, that is before 0.149 s. With the dual setting R2 trips in 0.060 s and R1 in 0.106 s, so the fuse is saved. With the forward setting alone R2 would need 0.160 s and the fuse would already be melting. Note that the fuse carries the grid current and the DG current together (5426 A), while each recloser only sees its own part.

The fault current at node 633 is 4.10 kA without the DG and 5.43 kA with it. At node 632 it is 4.74 kA and 6.53 kA.

Some more faults can be tried the same way, with the DG in. At the two-phase nodes choose the two-phase fault, at the single-phase nodes the line-to-ground fault.

Fault	Fuse	R1, fast	R2, fast	Fuse, melts
646, line-to-line	F646	2785 A, 0.189 s	1114 A, 0.086 s	3672 A, 0.415 s
684, line-to-line	F671-2	2543 A, 0.222 s	2543 A, 0.072 s	3963 A, 0.148 s
611, line-to-ground	F684	1814 A, 0.435 s	1718 A, 0.128 s	2236 A, 0.271 s

In all three cases both reclosers trip before the fuse melts.

6. Time-domain simulation (optional)

Import and activate `EMT_Fig10_DG_in.pfd` from the folder `2_EMT_Fig10_time_domain`. For the case without DG use `EMT_Fig10_DG_out.pfd`.

1. In the toolbar select the toolbox RMS/EMT Simulation.
2. Press Calculate Initial Conditions. Method: instantaneous values (EMT), step size 0.0001 s.
3. Press Start Simulation and give a stop time of 2 s.
4. Open the page Curve plot. If the plot looks empty, press the two scale-to-fit buttons.

A line-to-line fault at node 684 starts at 0.3 s and R2 opens and recloses twice. Without the DG the fuse melts at 1.248 s. With the DG it melts already at 0.752 s, in the second fast shot, so the fuse is not saved.

7. Running the Python scripts

The scripts repeat the complete study for all fault types at all nodes, with the DG out and in. This is what produces the table of 39 cases, which would take far too long by hand. They are started from PowerFactory with a Python Script object, which is an object that holds the name of a script file and has an Execute button.

7.1 Create the script object (once)

1. Activate the project and the study case as in section 3.
2. Open the Data Manager.
3. Look for an object called **Final(1)** in the library of the project (folder Scripts). If it exists, double-click it and use it. If it does not exist, select the Library folder of the project, click New Object, choose Python Script, name it `Run study` and press OK.
4. In the dialog, select the script file `install_packages.py` from `C:\DSDR\1_Main_DSDR_Fuse_Coordination\scripts`. Do not execute yet.

The dialog may look a little different in your version. It only has to run the file you choose when you press Execute. The messages of the script appear in the Output Window at the bottom of the screen.

7.2 Install the two packages (once)

With `install_packages.py` selected, press Execute. After a minute or so the Output Window should say "DONE. The packages are installed." If it says FAILED, check the internet connection and execute it again.

7.3 Run the study

1. In the same object change the script file to `show_progress.py` (same folder as above).
2. Check that the project and the study case Study with Substation Transformer are active.
3. Press Execute and wait. It takes a few minutes. First PowerFactory's own messages scroll by, and at the end the window is cleared and the report is printed.

The same report is saved in `scripts\results\Show_Progress_output.txt`. Messages like "Short-circuit calculation not possible" during the run are normal. Some nodes have only one or two phases, so not every fault type exists there (for example no three-phase fault at 645, 646, 684, 652 and 611), and the script simply goes on with the next case.

7.4 Check the result

Open `scripts\results\summary.txt`. It lists, out of 39 cases, how many keep the fuse saved:

Line in summary.txt	Cases
published fuse sizes, no DG	35 of 39
after steps 8–10, no DG (one fuse size changed)	39 of 39
DG connected, conventional recloser ("Fig. 14")	24 of 39
DG connected, revised fuses, conventional recloser ("check")	31 of 39
DG connected, dual-setting recloser and revised fuses ("Fig. 17")	38 of 39
same settings with the DG out of service	37 of 39

So without the DG almost all cases are fine, with the DG and a conventional recloser only 24 remain, and the dual-setting recloser together with the revised fuses brings it back to 38. The one case that stays lost is the line-to-ground fault at node 692.

The other results are in the same `results` folder: `Table_II.csv`, `Table_III.csv` and `Table_IV.csv` (branch currents, fuse coefficients, operating times), `figures` (the time–current plots Fig08 to Fig17 and Figures 1 to 6) and `database` (load flows, short circuits and Tables I to IV as spreadsheet files, with `Database_Report.txt`).

8. Problems you may meet

What you see	What to do
No script file field in the object, or the object cannot be created	Use **Final(1)** of the ready model, or create the object in the Library folder of the project with New Object, Python Script.
"Scripts folder not found"	The folder is not in `C:\DSDR\1_Main_DSDR_Fuse_Coordination`. Copy it there (section 2).
Installation FAILED, or "No module named matplotlib" (reportlab)	Check the internet connection and execute `install_packages.py` again.
Report ends with "SCRIPT STOPPED WITH AN ERROR"	Read the last lines. Usually the wrong study case is active, or a package is missing. Fix that and execute `show_progress.py` again.
The run takes more than 20 minutes	PowerFactory's own Python is slow for the calculation part. Just wait; the result is the same.
Fault currents far too high (6 to 8 kA at node 632 without DG)	The wrong study case is active. Choose Study with Substation Transformer.
R2 shows 470 A with the DG on	The machine is still out of service, or the load flow was not calculated again.
"Short-circuit calculation not possible" when done by hand	The node does not have that phase or fault type. Use node 632, 633 or 671.
The load flow does not converge	Press Reset Calculation, check that no switch was opened by hand, and calculate again.
Numbers differ in the last digit	This is normal.